Gyro Logic Stability Scene Tracing

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Gyro Logic Structure Slice Stability Core Core Gyro Logic Canonical Definition Gyro Logic

Canonical Core Slice local articulation Gyro realization

$$g_n = (S_n, B_n, c_n, \Sigma_n, a_n, K_n)$$

S_n Structure B_n Operator Orientation c_n Context Σ_n Slice process a_n local articulation K_n Stability Scene Core

$$S_n \xrightarrow{\Sigma_{B_n, c_n}} a_n \xrightarrow{\text{Stab}} K_n$$

`slice-done` underlying event `slice-done` Slice local establishment local unitization Operator Orientation Context protocol rule institutional criteria event-side transition underlying event process `slice-done` Stability global closure

Stability scalar equilibrium fixed point terminal condition articulation establishment residual local not-yet Stability Canonical Definition `readable` `Readable(...)` predicate readability domain-specific model semantics domain-relative placeholder

Incorporated Readability stored history distinction relation criterion relevance condition realization context update Continuity Readability Identity Trajectory state sequence chronological log Trajectory Gyro realization admissible relation contextual tracing Difference metric distance numerical error Boundary Slice Orientation Context codomain structured relation

schema relational structures graphs / hypergraphs order theory topology dynamical systems transition systems event structures category theory logic / proof theory constraint propagation probability / statistics sheaf-like structures process algebra partial model Gyro Logic

minimal theorem strict minimality formal commitment operational / exploratory unique minimality cardinal minimality order-theoretic minimality proof exploratory complete axiomatization universal semantics of readability universal Stability metric universal boundary-admissibility rule general tracing algorithm formal commitment validation comparison revision implementation study

: Gyro Logic Structure Slice Stability local articulation Incorporated Readability Continuity Readability contextual Trajectory Difference Boundary

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(1) `slice-done` local unitization (2) `Readable(...)` domain-relative placeholder (3) anti-post-hoc (4) retrospective establishment (5) `minimal` Gyro Logic Core Structure → Slice → Stability

Structure Slice Stability Stability Scene Trajectory Readable local articulation slice-done

1 Introduction

Gyro Logic Core

```
Structure
↓
Slice
↓
Stability
```

Gyro Logic Core Structure Slice Stability establishment Gyro Logic Canonical Definition Gyro Logic - formal organization

minimal schema uniquely minimal cardinally minimal formal ordering minimal minimum formal commitments minimality result admissible model class preservation criterion candidate model comparison relation component proof

Gyro Logic state-space model state space function domain codomain graph node edge dynamical trajectory ordered state sequence Stability equilibrium convergence fixed point robustness under perturbation scalar score Difference distance deviation error partial model Gyro Logic universal form

Slice Canonical Definition Slice Structure establishment path process Slice individuated Slice filtering projection selection ordinary retrieval Slice process process local articulation local articulation Stability

`slice-done` locality unfolding event process relation Operator local establishment `slice-done` underlying event intrinsic terminal state Orientation Con-

text inherited protocol rule institution frame Slice range articulated result status event-side transition boundary process

Stability Gyro Logic Stability evaluator decision-maker final completion numerical score fixed point Stability articulation establishment local scene scene confirmation continuation unresolved local not-yet local establishment residual not-yet Stability Scene Structure closure

Canonical Definition of Stability `readable` revision wording formal commitment universal independently validated `Readable(...)` operational semantics domain-specific model semantics readability articulation relation relevant frame established provisional explanatory / relational language

local realization continuation realization established realization event stored history append-only log established distinction relation criterion relevance condition context Incorporated Readability Continuity Readability Identity Trajectory chronological event list predefined state sequence Trajectory local Gyro realization admissible relation contextual tracing

Difference formal problem Gyro Logic Difference scalar metric symmetric error-like Orientation Context Slice partially defined relational ordered distributive field-like Boundary Difference Boundary Slice frame Difference articulated stabilized locally usable derivative distinction

final axiomatization exploratory Minimal Formal Model Structure Slice local articulation Stability Incorporated Readability Continuity Readability Trajectory Difference Boundary compact formal commitments notation supporting candidate Canonical Definition Core Gyro Logic relational structures graph theory topology dynamical systems category theory proof theory

Contribution Statement Research Questions Core formalization constraints Structure Slice Stability Incorporated Readability Continuity Readability contextual Trajectory Difference Boundary compact formal schema illustrative examples limitations

1.1 Contribution Statement

Gyro Logic Minimal Formal Model

Core Structure Slice Stability Canonical Definition Core provisional mathematical typing formal expression Canonical Definition definition formal candidate

unfolding process Slice process local articulation `slice-done` underlying event objective terminal state local unitization Slice pre-existing result extraction filtering selection universal stopping point

Stability scalar value equilibrium fixed point terminal condition articulation establishment structured local scene representation local establishment residual local not-yet Stability Scene Stability `slice-done`

Incorporated Readability stored history event log passive memory Incorporated Readability locally established distinction relation criterion relevance condition Gyro realization update accumulation addition revision integration reweighting invalidation loss of accessibility

Continuity Readability Identity Continuity admissible relation trace given Orientation Context Slice incorporated context continuity Identity break continuity Identity asserted continuity unavailable / disputed

Trajectory state sequence chronological log accumulated event Trajectory relation-bearing field event collection local Gyro realization admissible relation contextual tracing branching merging gap retrospective reinterpretation Re-Slice Jump Trajectory linear path

Difference metric distance numerical error Boundary Difference Slice Orientation Context structured relation of non-coincidence codomain scalar vectorial ordered relational distributive partially defined field-like Boundary Difference derivative locally usable distinction

model relational structures graphs / hypergraphs order theory topology dynamical systems transition systems event structures category theory logic / proof theory constraint propagation probability / statistics sheaf-like structures process algebra partial model assumption Gyro Logic

Core

```
Structure
↓
Slice
↓
Stability
```

exploratory integrated schema Gyro Logic model final canonical proven uniquely minimal theoretical distinction minimum formal commitments validation comparison revision implementation study

1.2 Research Questions

Research Question Gyro Logic Core object type premature reduction compact formal schema

RQ1. Structure Slice Stability Canonical meaning Core provisional mathematical type

RQ2. Slice result path fully individuated `slice-done` underlying event objective end local articulation unfolding process

RQ3. Stability `slice-done` local unitization residual local not-yet locally established and continuable scene

RQ4. local Gyro realization established stored history passive memory monotonic accumulation realization

RQ5. Continuity Trajectory Identity predefined state sequence chronological log universal `Readable(...)` predicate admissible relation contextual tracing

RQ6. schema partial model assumption Gyro Logic restrictive

scope Gyro Logic axiomatize discipline reduce theory semantics admissibility conditions internally consistent explicitly provisional - formal organization

2 The Invariant Core and Formalization Constraints

2.1 Core

formal model Gyro Logic Core

```
Structure
↓
Slice
↓
Stability
```

Core order composition variable concept Core derivative concept Core Orientation Context local articulation Incorporated Readability Continuity Readability Trajectory Difference Boundary Operator Response Re-Slice Jump conditioning resulting relational temporal interpretive concept local Gyro realization formal description Core

Canonical Definitions

> **Structure is the mode in which something can be established.**
>
> **Slice is the process by which a path is opened through a Structure toward an establishment.**
>
> **Stability is the state in which an opened path becomes readable as an establishment that can continue.**

mathematical expression Canonical Definition formal candidate Canonical Definition formal candidate object Canonical Definition

Stability Definition `readable` canonical wording wording universal formal predicate `Readable(...)` formal readability notation specialized model semantics provisional domain-relative placeholder

2.2 Canonical Definition Formal Candidate

statement level Canonical Definition Gyro Logic concept theoretical meaning Formal Candidate meaning provisional mathematical organization

```
canonical definition
≠
formal candidate
```

formula replacement definition disciplined representational proposal Structure S_n Slice process Σ_n Stability Scene K_n Structure set element Slice ordinary total function Stability tuple

notation silent ontological commitment function fixed domain / codomain graph preindividuated node / edge metric numerical comparability symmetry triangle inequality state trajectory already defined state space temporal ordering formal model notation commit open

2.3 Minimal Formal Commitments

minimum commitments

analysis local Gyro realization reality intrinsically independent units local realization reference realization relation

Slice Slice local articulation process locally available articulation

Stability Slice process local articulation Stability appearance articulation establishment continuation

local realization established readability realization condition conditioning deterministic monotonic complete immediately adjacent in time

local realization relation Orientation Context Slice continuity relation existence traceability admissibility continuity judgment

Difference universally scalar metric symmetric total error-like

commitment minimal schema Structure relation field context update tracing operation mathematical type later specialization open

2.4 Formalization Constraints

candidate model constraints acceptable

Core preservation. Structure Slice Stability order composition replacement Core

Definition preservation. Canonical concept narrow mathematical special case

Process–result separation. Slice unfolding process local articulation

Articulation–Stability separation. local articulation established and continuable Stability

Locality without global closure. locally established Stability Scene globally open Structure residual local not-yet

Non-reductive readability update. Incorporated Readability append-only history immutable stored data

Identity–continuity separation. continuity without identity identity claim without available continuity

Trajectory–sequence separation. Trajectory chronological log event set predefined linear state sequence

Difference–metric separation. Difference metric / error-model assumptions

Layer consistency. Gyro Logic theory model GyroOS implementation decision GyroAuth application requirement theory concept

2.5 Explicit Non-Assumptions

Minimal Formal Model Structure fixed mathematical object type relevant object / state / relation / boundary Slice individuated Slice deterministic / total function Stability scalar threshold / equilibrium / fixed point readability monotonically accumulates continuity identity Trajectory linear Difference distance Gyro Logic complete foundation

construction status non-assumption metric graph dynamical system category proof context transition system assumption domain valid instantiation theory universal form

3 Structure as Establishability Without Fixed Mathematical Type

3.1 Canonical Meaning and Formal Problem

Structure Canonical Definition

Structure is the mode in which something can be established.

definition Structure state object set space relation container substrate configuration - particular domain valid representation universal mathematical type formal problem Structure object type specialization commit

mathematical modeling object state variable relation individuated Gyro Logic relevant object / relation fixed Slice locally articulable prior formal condition Structure completed inventory of established entities establishability-bearing organization working characterization replacement definition

3.2 Structure Is Not the Current State

current state Structure state establishment Structure current state x_n minimal model

$$S_n = x_n$$

state Structure relative establishable

$$x_n \lhd S_n$$

$\lhd$ provisional establishment-availability relation appropriate conditions x_n established / articulable / available domain-specific model set membership physical containment logical entailment part–whole inclusion

Structure state available current state Structure object apparent state Structure

3.3 Structure Is Not the Bearer or Object

local realization entity material system text institution process bearer bearer Structure cake software system legal institution authentication session example bearer Structure distinction relation state possible establishment available mode

```
bearer
=
Structure
=
current state
```

collapse bearer different conditions multiple Structures support multiple bearers one relational Structure participate bearer persist current state Structure bearer material / descriptive organization continue model

3.4 Structure as Globally Not-Yet

Slice Structure global not-yet characterize absence nothingness ignorance empty possibility set prospective Slice articulate particular local establishment form available

$\mathcal{A}^*(S_n)$ Structure available articulation family provisional placeholder asterisk fully individuated object commit

$$a \in \mathcal{A}^*(S_n)$$

ordinary set membership appropriate Slice a S_n compatible / supportable / realizable

$\mathcal{A}^*(S_n)$ pre-existing answer catalogue Slice establishability under-determined placeholder all candidate articulation enumerable mutually exclusive simultaneously available Orientation / Context invariant

3.5 Minimal Relational Characterization

Structure

$$S_n = \langle \mathsf{Avail}_n, \mathsf{Rel}_n, \mathsf{Cond}_n \rangle^*$$

reference

- Avail_n local establishment available
- Rel_n establishment support / constrain / connect relation
- Cond_n availability / relation relevant condition
- * universal tuple ontology

schema state space graph constraint system topological space component complete explicit independent directly observable Slice local articulation availability / relation / conditioning Structure support

relation

$$\mathsf{Establishable}(a; S_n, B_n, c_n)$$

articulation a Structure S_n Orientation B_n Context c_n relative locally available a Slice already given appearance Stable

3.6 Orientation and Context Do Not Constitute Structure

Orientation Context Structure aspect Slice relevant condition Structure Operator Structure pe relative representation collapse representation constrain / resist / exceed

$$S_n \neq S_n(B_n, c_n)$$

Structure Orientation-conditioned view warning specialized model

$$\mathrm{Pres}_{B_n, c_n}(S_n)$$

accessible presentation S_n exhaust

3.7 Local Establishment Does Not Close Structure

Slice local articulation yield Stable resulting local establishment Structure globally close

$$S_n \overset{\Sigma_{B_n, c_n}}{\Longrightarrow} a_n$$

Stability Scene K_n available

$$\mathcal{A}^*(S_n) = \{a_n\}$$

S_n is complete

local realization articulation settle relation distinction possibility unresolved

3.8 Formal Commitments and Non-Commitments

Structure component Structure locally reference establishment possibility support current state bearer articulation available constrain one local establishment open - commit

Structure set manifold category graph state space probability space constraint system logical theory physical substrate commit directly observable fully enumerable temporally static internally homogeneous prior incorporated readability independent

4 Slice as Process and Local Articulation

4.1 Canonical Meaning

Slice Canonical Definition

> **Slice is the process by which a path is opened through a Structure toward an establishment.**

definition constraints Slice completed object process establishment path process open fully individuated entity

Slice fixed result space operation filtering projection selection retrieval partitioning ordinary extraction restricted domain Slice process instantiate Slice universal meaning

4.2 Why Extraction Models Are Insufficient

extraction model schematically

$$E : S \to X$$

model result type already known domain Gyro Logic assumption output already individuated codomain fixed relevant distinction available operation pre-existing component reveal

Gyro Logic some Slice processes extraction implemented extraction Slice theoretical meaning general Slice locally available form process constituted

```
Slice
≠
extraction of an already completed result

path-opening
≠
retrieval of a pre-existing path object
```

4.3 Process and Local Articulation

local Structure S_n Operator Orientation B_n Context c_n Slice process Σ_n provisional relation

$$S_n \overset{\Sigma_{B_n, c_n}}{\Longrightarrow} a_n$$

a_n Slice available local articulation

$\Rightarrow$ ordinary total function result partial context-dependent non-deterministic retrospectively readable another Orientation unavailable process relation

1. local Structure
2. Slice Orientation Context unfold
3. local articulation unfolding available
4. articulation process

a_n final completion global closure Stability established / continuable evaluate locally available form

```
Slice process
≠
local articulation
```

```
local articulation
≠
Stability
```

4.4 Slice-ing and Slice-done

Gyro Logic Slice time-including unfolding unfolding result unitize

```
slice-ing
=
the process while Slice is unfolding
```

```
slice-done
=
a local unitization in which some range of the unfolding Slice
is treated as one local establishment
```

local unitization current Operator judgment Orientation and Context inherited protocol / rule institutional criteria strong event-side transitions underlying event point objectively / absolutely end

```
slice-done
≠
end of the underlying event
```

```
slice-done
≠
Stability
```

provisional process representation

$$\alpha_\Sigma : I_\Sigma \to \mathcal{A}^*(S_n)$$

$$a_n = \alpha_\Sigma(\tau^*)$$

I_Σ internal process index $\mathcal{A}^*(S_n)$ possible local articulation provisional space τ^* selected frame local analytical point of unitization notation illustrative canonical physical time unique terminal index fixed articulation space universally

general

$$(S_n, B_n, c_n, \Sigma_n; F_n) \rightsquigarrow a_n$$

F_n domain model local / inherited frame F_n additional Core element

4.5 The Role of Orientation, Context, and Inherited Frames

Orientation Context Slice condition additional Core element Orientation Structure directional entrance Context relation distinction articulation available surrounding condition

local boundary current Operator freshly choose protocol termination rule legal / institutional criterion medical procedure pre-existing computational contract inherit event-side transition available boundary strongly constrain inherited criterion selected frame boundary locally fix possible source / constraint boundary universal claim

$$\Sigma_{B_n, c_n}$$

indexed notation conditioning Structure Operator Structure single observer relative

4.6 Minimal Anti-Post-Hoc Constraint

Operator-relativity inherited-frame language after-the-fact rescue mechanism

> boundary judgment Orientation Context inherited rule institutional criterion boundary provenance boundary judgment support boundary frame boundary

Temporal priority prior statement all plausible candidate boundaries open constraint after the fact inherited label non-post-hoc claimed provenance independently supportable

frame specific / discriminating / evidentially sufficient domain-neutral metric never-live alternative exclude substantive constraint full admissibility semantics future work

4.7 Slice Does Not Consume Structure

Slice process Structure exhausted / consumed foreground background highly determinate articulation Structure relation possible articulation unresolved condition alternative path

```
Structure after Slice
≠
Structure minus extracted result
```

Slice later approach condition literal subtraction all Structure change Slice external interaction environmental change material transformation later realization condition

4.8 Locality and Non-Closure

local articulation Structure section Orientation / Context inherited frame particular establishment path local Structure remainder irrelevant / nonexistent

$$a_n \text{ is available}$$

$$S_n \text{ remains globally open}$$

Re-Slice alternative articulation Context expansion Difference recognition later Trajectory tracing

4.9 Minimal Formal Commitments for Slice

Slice commit

1. Slice processual static mapping result
2. process local articulation
3. `slice-done` local unitization universal event termination
4. local articulation fully individuated object
5. Orientation Context inherited frame Slice constrain Core stage
6. local articulation appearance / unitization Stability
7. Slice Structure consume / close
8. extraction projection filtering classification selection domain-specific implementation Slice universal definition

4.10 Explicit Non-Commitments

every Slice unique result every Slice terminates Slice deterministic articulation space - fixed Orientation human observer Context fully representable every local boundary current Operator freely chosen `slice-done` irreversible

a path is opened literal geometric path path relational logical procedural semantic causal material institutional domain-specific

4.11 Transition to Stability

local articulation a_n next formal distinction result Stability Scene articulation Stability transition articulation relevant Structure Orientation Context local / inherited frame establishment

$$K_n = \mathsf{StabScene}(a_n; S_n, B_n, c_n, F_n)$$

optional F_n domain-model parameter Core element

5 Stability as a Readable and Continuable Scene

5.1 Canonical Meaning

Stability Canonical Definition

> **Stability is the state in which an opened path becomes readable as an establishment that can continue.**

definition Stability Slice Slice completion Slice path open `slice-done` local unitization Stability articulation establishment

```
Slice process
≠
local articulation / local unitization
≠
Stability
```

local articulation appearance / unitization domain-relevant continuation conditions Stability relevant frame articulation establishment continuation

5.2 Why a Scalar Is Not Sufficient

implementation-oriented setting Stability score threshold probability confidence value robustness measure operational indicator theoretical Stability

$$\sigma_n \in [0, 1]$$

scalar specific model degree of assessed stability which relations are available for establishment which unresolved conditions remain which continuations are available

```
Stability score
≠
Stability
```

$$\sigma_n \geq \theta$$

threshold particular implementation policy labeling Canonical concept

5.3 Why Equilibrium and Fixed Points Are Partial Models

Equilibrium convergence invariant set attractor fixed point dynamical systems powerful stability models relevant state space dynamics perturbation model application Gyro Logic instantiate

Gyro Stability motionless globally converged invariant terminal locally established scene change continuation coherent long-run limit establishment available

```
Gyro Stability
≠
equilibrium only
≠
fixed point only
≠
global convergence only
```

5.4 Stability as a Structured Local Scene

Stability Scene provisionally

$$K_n = (a_n, L_n, U_n, C_n^+)$$

- a_n Slice available local articulation
- L_n scene articulation established current relation / distinction / condition
- U_n unresolved / unavailable residual local not-yet
- C_n^+ scene support continuation condition / available continuation

tuple formal candidate replacement definition value representation

$$K_n = \text{StabScene}(a_n; S_n, B_n, c_n)$$

Stability Structure Orientation Context relative evaluated deterministic total single predicate reducible

5.5 Readability and Continuability

Canonical Definition `readable` word universal necessary-and-sufficient predicate semantics

weak domain-relative formulation

$$\text{Stable}(a_n; S_n, B_n, c_n) \Rightarrow \text{EstablishedFor}_D(a_n; S_n, B_n, c_n) \wedge \text{Continuable}_D(a_n; S_n, B_n, c_n)$$

subscript D domain-specific interpretation specialized model establishment condition Readable_D - domain model placeholder establish universal Gyro predicate

implication articulation established continuation support Stability claim converse universally

```
continuable
≠
unchanging
```

```
readable / established-for
≠
final
```

5.6 Residual Not-Yet

Stability unresolved local not-yet U_n

```
local establishment
+
residual local not-yet
```

scene Stability Structure closure

$$U_n \neq \emptyset \quad \text{is compatible with} \quad K_n \text{ being stable}$$

every Stability Scene unresolved elements formal model $U_n = \emptyset$

5.7 Locality and Neighborhood Interpretation

Stability isolated point local scene / neighborhood neighborhood structure application

$$K_n \subseteq N(a_n)$$

$N(a_n)$ admissible range of variation articulation established / continuable neighborhood

topology universal foundation neighborhood topological relational semantic operational probabilistic domain-specific

5.8 Stability Does Not Decide

Stability evaluated evaluate Continue Stop Jump Re-Slice Defer response select Core - operational extension Operator Response

```
Stability
≠
Operator Response
```

```
Structure
→ Slice
→ Stability
→ Operator Response
```

arrow Gyro Process Core

5.9 Stability and Later Structure

Stability Scene later Structure available unchanged transfer established distinction relation continuation condition later context incorporated revised weighted invalidated inaccessible

$$K_n \rightsquigarrow q_n \rightsquigarrow \Gamma_{n+1}$$

q_n local realization incorporated Γ_{n+1} later readability context

5.10 Minimum Formal Commitments

Stability model commit

1. Stability Slice local articulation
2. Stability local establishment continuation support
3. Stability single scalar internal structure
4. Stability residual local not-yet
5. Stability local Structure globally close
6. Stability operational decision
7. Stability Scene Incorporated Readability later realization condition

5.11 Transition to Incorporated Readability

articulation Stability Scene established / continuable readability later realization available persist entire event / state / scene immutable record stored Incorporated Readability simple history preservation context update

6 Incorporated Readability and Context Update

6.1 From Local Stability to Later Conditions

local Gyro realization isolated Stability Scene articulation establishment readability later realization available Incorporated Readability

Incorporated Readability preceding event Slice process articulation Stability Scene local realization later condition usable established distinction relation criterion relevance ordering Boundary Difference pattern continuity condition later Orientation tendency

$$g_n = (S_n, B_n, c_n, \Sigma_n, a_n, K_n)$$

incorporated readability

$$q_n = \mathrm{Inc}(g_n)$$

Inc complete and lossless summary deterministically extract operator g_n available readability later conditions available provisional relation

6.2 Incorporated Readability Is Not Stored History

stored history something occurred record Incorporated Readability subsequent establishment available

```
history of prior realization
≠
readability available to later realization
```

log event later interpretation original event explicit record incorporated distinction later interpretation

archive H_n readability context Γ_n

$$H_{n+1} = \text{Append}(H_n, g_n)$$

$$\Gamma_{n+1} = \text{Update}_\Gamma(\Gamma_n, q_n)$$

occurrence record later reading comparison orientation establishment conditions

6.3 Context as an Available Readability Condition

Γ_n provisional readability context every domain fixed proposition set contain / organize

- now available distinctions
- now followable relations
- now applicable criteria
- prior Difference patterns affecting comparison
- usable Boundaries
- relevance weights / priority orderings
- exclusions / invalidations / unresolved conflicts
- conditions under which later Slice may proceed

weak characterization

$$\Gamma_n = \langle \mathsf{Avail}_n, \mathsf{Weight}_n, \mathsf{Constraint}_n, \mathsf{Access}_n \rangle^*$$

* canonical definition formal candidate

6.4 Non-Monotonic Update

Incorporated Readability monotonically grow later update addition revision integration reweighting invalidation suppression loss of accessibility

$$\Gamma_n \not\subseteq \Gamma_{n+1}$$

$$\Gamma_{n+1} = \Gamma_n \cup \{q_n\}$$

$$\Gamma_{n+1} = \mathrm{Update}_{\Gamma}(\Gamma_n, q_n, e_n)$$

e_n local Gyro realization environmental / institutional / interpersonal / material change

```
Structure change
≠
Slice
```

all later conditions preceding Slice produced claim

6.5 Weighted Incorporated Readability

incorporated readability influence distinction available peripheral distinction later Context decisive

$$w_n(q; c, B) \in W$$

W numeric ordering priority class partial order

$$\mathrm{Effective}(q_n; B_m, c_m, \Sigma_m)$$

one later realization hold another fail Incorporated Readability permanent universal rule context-relative influence later Structure conditions woven readability

6.6 Structure Update

later Structure completely independent object Incorporated Readability derived

$$(S_n, \Gamma_{n+1}, e_n) \rightsquigarrow S_{n+1}$$

relation update partial distributed non-deterministic only retrospectively readable

```
Structure
≠
readability context
```

incorporated readability later Structure conditions

6.7 Example: Mathematical Reasoning

mathematical problem definition lemma intermediate equality admissible transformation final proof established established later step step history later reasoning legitimately use

$$\Gamma_{n+1} = \mathrm{Update}_{\Gamma}(\Gamma_n, q_n)$$

later correction q_n revise / invalidate effective context model incorporation retraction support

6.8 Minimal Commitments

1. local Gyro realization some readability later realization available
2. incorporated complete prior realization
3. incorporated readability stored history later conditions
4. update non-monotonic context-relative
5. external change local Gyro realization change formal

Γ_n logical theory database memory store vector state probability distribution

6.9 Transition to Continuity Readability

Incorporated Readability local establishment later conditions available two local realizations connected relation existence traceability admissibility current continuity judgment

7 Continuity Readability and Identity

7.1 From Local Establishment to Relational Continuity

preceding section context update later comparison / tracing two realizations continuous relation untraceable traceable given Orientation Context Slice incorporated context continuity count

```
relation existence
≠
traceability
≠
admissibility
≠
continuity judgment
```

Continuity two realizations intrinsically property specific conditions available relational judgment

7.2 Local Gyro Realizations

$$g_i = (S_i, B_i, c_i, \Sigma_i, a_i, K_i)$$

component Structure Operator Orientation Context Slice process local articulation Stability Scene tuple bookkeeping schema every realization ontologically six independent objects

7.3 Relation Existence

candidate relation r

$$r(g_i, g_j)$$

$$g_i \xrightarrow{r} g_j$$

r domain-relative causal succession functional succession semantic inheritance material transfer recognized Difference pattern Boundary correspondence response-to-orientation linkage retained readability condition institutional / rule-based connection

7.4 Traceability

$$\text{Traceable}(g_i, g_j; r)$$

relation one realization another follow available evidence incorporated readability temporal reach admissible inference rule access condition

$$r(g_i, g_j) \nRightarrow \text{Traceable}(g_i, g_j; r)$$

relation existence traceability

7.5 Admissibility

traceable relation relevant continuity relation

$$\text{Adm}(r; B, c, \Sigma, \Gamma)$$

Orientation Context continuity-oriented Slice incorporated readability context r admissible predicate

admissibility relevance scope permitted inference causal sufficiency semantic compatibility material continuity institutional validity temporal accessibility trust / evidence requirement encode universal static binary

7.6 Continuity Readability

Continuity Readability Gyro term universal independent `Readable(...)` predicate

$$\text{CR}(g_i, g_j; B, c, \Sigma, \Gamma) \iff \exists r(\text{Adm}(r; B, c, \Sigma, \Gamma) \wedge \text{Traceable}(g_i, g_j; r) \wedge \text{CountsAsContinuity}_D(r; B, c, \Sigma, \Gamma))$$

$\text{CountsAsContinuity}_D$ explicitly domain-relative placeholder specialized model justified Readable_D graded evidence structure logical rule causal criterion institutional rule

```
Is there an admissible relation?
Can it be traced?
Does the domain model currently count it as continuity?
```

operational question

7.7 Continuity Readability Is Context-Relative

same pair of realizations

$$\text{CR}(g_i, g_j; B_1, c_1, \Sigma_1, \Gamma_1) \neq \text{CR}(g_i, g_j; B_2, c_2, \Sigma_2, \Gamma_2)$$

continuity arbitrary admissibility domain-relative continuity condition explicit conditions depend

7.8 Identity as a Separate Criterion

$$\mathrm{Id}_q(g_i, g_j)$$

identity criterion q g_i, g_j same entity / bearer / Structure q numerical legal functional material persistence semantic account biological role identity

$$\mathrm{CR}(g_i, g_j) \not\equiv \mathrm{Id}_q(g_i, g_j)$$

Continuity admissible relation trace continuity count Identity criterion same different questions

7.9 Continuity Without Identity

$$\mathrm{CR}(g_i, g_j) = \text{true}$$

$$\mathrm{Id}_q(g_i, g_j) = \text{false}$$

batter cake type / object criterion different material transformation causal succession production history continuity

```
Identity break
≠
Trajectory break
```

7.10 Identity Without Readable Continuity

$$\mathrm{Id}_q(g_i, g_j) = \text{true}$$

$$\mathrm{CR}(g_i, g_j) = \text{false}$$

indeterminate institution same legal person assert continuity reconstruction system same account identifier behavioral / operational trajectory unavailable

7.11 Continuity Readability and Difference

Continuity Difference absence structured Difference pattern trace continuity available

$$\Delta_{B,c,\Sigma}(g_i, g_j) \neq 0$$

Difference continuity relation admissible continuity

```
continuity
≠
unchanged sameness
```

7.12 Continuity Readability and Incorporated Readability

Γ relation admissible continuity rule apply

$$\Gamma_{n+1} = \mathrm{Update}_{\Gamma}(\Gamma_n, q_n, e_n)$$

$$\mathrm{CR}(g_i, g_j; B, c, \Sigma, \Gamma_{n+1})$$

Γ_n judgment previously unavailable relation later evidence traceable / continuity-counting previously accepted continuity invalidated

7.13 Binary and Graded Forms

$$\mathrm{CR}^*(g_i, g_j; B, c, \Sigma, \Gamma) \in \mathcal{C}$$

$\mathcal{C}$ ordered set confidence interval evidence structure domain-specific classification Continuity universally binary / numerical

7.14 Minimal Commitments

1. Local Gyro realization relation
2. Relation existence traceability admissibility continuity judgment
3. Continuity Readability Orientation Context Slice incorporated readability depend
4. Identity separate criterion
5. Continuity Difference identity change persist
6. Identity continuity unavailable / disputed assert
7. Continuity reading Re-Slice / context update revise

Continuity equivalence relation always transitive symmetric globally decidable

7.15 Transition to Contextual Trajectory

particular local realizations connected broader Trajectory Trajectory family of local realizations retained relation field contextual tracing operation

8 Contextual Trajectory

8.1 From Local Continuity to Trajectory

Trajectory additional Core element multiple local Gyro realization given Orientation Context Slice incorporated readability condition connected derivative relational construction

```
Trajectory
≠
state sequence
≠
chronological log
≠
event collection
≠
relation-bearing field itself
```

8.2 Local Gyro Realizations

$$g_i = (S_i, B_i, c_i, \Sigma_i, a_i, K_i)$$

family

$$G = \{g_i\}_{i \in I}$$

G Trajectory one or more trajectories support local realizations collection

8.3 The Relation-Bearing Trace Field

possible relation type family $\mathcal{R}$

$$E \subseteq G \times \mathcal{R} \times G$$

$$\mathcal{G}_R = (G, E)$$

relation-bearing trace field $(g_i, r, g_j) \in E$ type r relation available / retained / inferred / representable causal material functional semantic procedural institutional identity-related Boundary-related Difference-related

```
relation-bearing trace field
≠
Trajectory
```

8.4 Contextual Tracing

Trajectory

$$T_{B,c,\Sigma_T,\Gamma_T} = \mathrm{Trace}_{B,c,\Sigma_T,\Gamma_T}(G, E)$$

provisionally tracing operation G E enumerate current conditions admissible traces select compose suppress weight interpret

same relation-bearing field different Orientation / Context / Slice / readability context different Trajectory

8.5 Admissibility of a Trace

candidate trace

$$\pi = (g_{i_0}, r_1, g_{i_1}, r_2, \ldots, r_m, g_{i_m})$$

Trajectory inclusion formal adjacency

$$\mathrm{AdmTrace}(\pi; B, c, \Sigma_T, \Gamma_T)$$

relation type admissibility successive relation compatibility relevance weight retained Difference pattern Boundary condition continuity criteria missing intermediate realizations contextual interpretation constraints depend

8.6 Trajectory Is Not a Predefined State Sequence

ordinary state trajectory

$$x_0, x_1, x_2, \dots, x_n$$

common state space ordering relation presuppose Gyro Trajectory weaker assumptions connected realizations type representation granularity identity heterogeneous relations continuity

8.7 Trajectory Is Not a Log

chronological log event order stored record relation admissible / traceable / continuity - count

$$H \neq T_{B,c,\Sigma_T,\Gamma_T}$$

same history multiple trajectories no current trajectory later trajectory support

8.8 Branching, Merging, and Multiple Trajectories

Trajectory linear branching merging parallel trajectories competing trajectories nested trajectories partial trajectories implementation graph-like hypergraph-like partially ordered category-like event-structural universal form

8.9 Gaps and Unavailable Intervals

missing intermediate realization Trajectory break gap admissible relation trace continuity available dense record admissible continuity relation Trajectory

```
record gap
≠
Trajectory break
```

```
dense history
≠
continuity
```

8.10 Retrospective Tracing and Re-Slice

later realization previously unavailable relation / evidence Re-Slice earlier realizations retained relations different way trace

$$T^{(n)} = \mathrm{Trace}_{B_n,c_n,\Sigma_n,\Gamma_n}(G, E)$$

$$T^{(n+1)} = \mathrm{Trace}_{B_{n+1},c_{n+1},\Sigma_{n+1},\Gamma_{n+1}}(G, E)$$

change past altered retained traces present organization later conditions

related but distinct problem retrospective establishment later Operator available trace relation consequence later establishment earlier event present local establishment

```
past event itself
≠
present establishment about the past event
```

one remaining trace past event uniquely determine

8.11 Jump and Non-Continuous Reconstruction

Jump large numerical discontinuity current continuity conditions existing trace organization admissible continuation support reconstruction

```
Jump
≠
large Difference only
```

```
Jump
≠
necessary deletion of prior Trajectory
```

8.12 Relation to Incorporated Readability

Γ_T change previously unavailable relation expose accepted relation invalidate competing traces weight change separated realizations connect one Trajectory split multiple trajectories merge

Trajectory prior Gyro realization independent stored accumulation reducible

8.13 Minimal Commitments

1. local Gyro realization provisionally reference
2. heterogeneous relations represent
3. relation existence traceability admissibility continuity judgment
4. tracing operation Orientation Context Slice Incorporated Readability condition
5. tracing result non-linear partial revisable plural
6. Trajectory derivative Core

8.14 Transition to Difference and Boundary

Tracing local realization possible relation distinction Difference metric distance / error Boundary Difference

9 Difference and Boundary

Difference Boundary derivative concepts Core replace Structure Slice Stability additional stage Orientation Context Slice non-coincidence available / organized / usable

9.1 Difference Is Not Distance

Difference numerical distance deviation residual error metric space norm comparison scale target value domain universally

$$\Delta_{B,c,\Sigma} : X \rightharpoonup D$$

D scalar vector ordered tuple relation partial order distribution symbolic classification field-like object partial arrow every input Difference available

metric

$$d : X \times X \to \mathbb{R}_{\geq 0}$$

additional commitments special case

```
Difference
≠
Distance
```

9.2 Difference Is Not Error

Error reference norm target expected value accepted state presuppose Difference evaluative meaning

```
Difference
≠
Error
```

change in form / role / interpretation / relation / continuation failure Difference

9.3 Difference as Slice-Relative Structured Non-Coincidence

working characterization

> Difference is a Slice-relative structured relation of non-coincidence.

Difference Slice-relative structured relational pairwise

$$\Delta_{B,c,\Sigma}(x, y)$$

general

$$\Delta_{B,c,\Sigma}(X)$$

9.4 Difference and Local Articulation

Difference Slice already usable object Slice local articulation a_n structured non-coincidence available

$$S_n \xRightarrow{\Sigma_{B_n,c_n}} a_n$$

Difference pattern Stability Scene L_n U_n C_n^+ contribute Stability Difference zero

$$\Delta_n \neq 0$$

Stability compatible $\Delta_n = 0$ Stability / Identity / Continuity Readability

9.5 Boundary Is Not Difference

Boundary Difference Difference structured non-coincidence Boundary particular Slice locally usable distinction

```
Difference
→ may become locally usable as a distinction
→ Boundary
```

mandatory temporal sequence derivative relation

`Difference`

≠

`Boundary`

9.6 Boundary as a Slice-Relative Distinction

supporting characterization

Boundary is a Slice-relative distinction that has become readable through Slice.

`readable` canonical explanatory role universal independent predicate Boundary fixed line intrinsically contained in Structure spatial logical semantic operational social temporal procedural hybrid

$$\mathrm{Bd}_{B,c,\Sigma,\Gamma}(d)$$

distinction d stated conditions Boundary function

9.7 Boundary State

$$\mathrm{BS}(x \mid d, B, c, \Sigma, \Gamma)$$

object / event / articulation / realization Boundary d relative state normal non- unabsent blank unknown void-relative inside outside crossing deferred domain-specific classification

9.8 Boundary, Continuity, and Trajectory

Boundary one kind of continuity interrupt another preserve type Boundary crossing Identity criterion break material / causal / semantic / functional continuity available

`Boundary crossing`

≠

`Trajectory break`

9.9 Incorporation of Difference and Boundary Readability

$$q_n^{\Delta} = \mathrm{Inc}_{\Delta}(g_n)$$

$$q_n^{\mathrm{Bd}} = \mathrm{Inc}_{\mathrm{Bd}}(g_n)$$

later context update

$$\Gamma_{n+1} = \mathrm{Update}_{\Gamma}(\Gamma_n, q_n^{\Delta}, q_n^{\mathrm{Bd}}, e_n)$$

inclusion permanent preservation revision reweighting invalidation suppression loss of accessibility

9.10 Formal Commitments and Non-Commitments

1. Difference Orientation Context Slice relative
2. Difference partial / heterogeneous
3. Difference universally metric / error-like
4. Boundary locally usable distinction derivative Difference
5. Boundary State relational / provisional
6. Difference Boundary Stability Continuity Readability Trajectory later incorporated readability

10 Minimal Formal Model

10.1 Purpose of the Integrated Schema

preceding sections components minimal schema integrate complete axiomatization final mathematical ontology current theoretical distinctions smallest set of distinguishable objects / relations

Core

```
Structure
↓
Slice
↓
Stability
```

local realization conditions Slice available articulation later context incorporated readability continuity / Trajectory relations represent

10.2 Local Gyro Realization

$$g_n = (S_n, B_n, c_n, \Sigma_n, a_n, K_n)$$

component Structure Operator Orientation Context Slice process local articulation Stability Scene tuple representational convenience Orientation Context local articulation Core

$$S_n \xrightarrow{\Sigma_{B_n,c_n}} a_n \xrightarrow{\text{Stab}} K_n$$

first relation Orientation / Context Slice process second relation local articulation Stability Scene transition deterministic total function

10.3 Structure

Structure S_n reference mathematical type open

$$x \lhd S_n$$

x S_n relative locally establishable / available weak relation

10.4 Slice and Local Articulation

$$S_n \xrightarrow{\Sigma_{B_n,c_n}} a_n$$

```
Slice process
≠
local articulation
```

a_n Slice fully individuated result Stable

10.5 Stability Scene

$$K_n = (a_n, L_n, U_n, C_n^+)$$

provisionally $U_n \neq \emptyset$ Stability Scene

$$\mathrm{StabScene}(a_n; S_n, B_n, c_n)$$

articulation relevant conditions established / continuable weak condition

10.6 Incorporated Readability

$$q_n = \mathrm{Inc}(g_n)$$

$$\Gamma_{n+1} = \mathrm{Update}_{\Gamma}(\Gamma_n, q_n, e_n)$$

append-only history

$$(S_n, \Gamma_{n+1}, e_n) \rightsquigarrow S_{n+1}$$

all Structure change preceding Gyro realization generated

10.7 Continuity Readability

$$\mathrm{CR}(g_i, g_j; B, c, \Sigma, \Gamma) \iff \exists r(\mathrm{Adm}(r) \wedge \mathrm{Traceable}(r) \wedge \mathrm{CountsAsContinuity}_D(r))$$

weakly

```
relation existence
≠
traceability
≠
admissibility
≠
continuity judgment
```

Identity $\mathrm{Id}_q(g_i, g_j)$ separate relation

10.8 Relation-Bearing Trace Field and Trajectory

$$G = \{g_i\}_{i \in I}$$

$$E \subseteq G \times \mathcal{R} \times G$$

$$\mathcal{G}_R = (G, E)$$

trace field Trajectory

$$T_{B,c,\Sigma_T,\Gamma_T} = \mathrm{Trace}_{B,c,\Sigma_T,\Gamma_T}(G,E)$$

Trajectory contextual tracing result branching merging gap retrospective reinterpretation Re-Slice Jump

10.9 Difference

$$\Delta_{B,c,\Sigma} : X \rightharpoonup D$$

D heterogeneous

```
Difference
≠
Distance
≠
Error
≠
Boundary
```

10.10 Compact Integrated Form

$$g_n = (S_n, B_n, c_n, \Sigma_n, a_n, K_n)$$

$$S_n \overset{\Sigma_{B_n,c_n}}{\Longrightarrow} a_n$$

$$K_n = \mathrm{StabScene}(a_n; S_n, B_n, c_n)$$

$$q_n = \mathrm{Inc}(g_n)$$

$$\Gamma_{n+1} = \mathrm{Update}_{\Gamma}(\Gamma_n, q_n, e_n)$$

$$(S_n, \Gamma_{n+1}, e_n) \rightsquigarrow S_{n+1}$$

$$\mathrm{CR}(g_i, g_j) \iff \exists r : \mathrm{Adm}(r) \wedge \mathrm{Traceable}(r) \wedge \mathrm{CountsAsContinuity}_D(r)$$

$$\mathcal{G}_R = (G, E)$$

$$T = \mathrm{Trace}(G, E)$$

$$\Delta_{B,c,\Sigma} : X \rightharpoonup D$$

10.11 What the Model Guarantees

current exploratory level model guarantee conceptual / formal separation Structure Slice process local articulation Stability Scene history Incorporated Readability relation existence Continuity Readability Identity continuity trace field Trajectory Difference metric / error / Boundary Gyro Logic / GyroOS / GyroAuth layer separation

10.12 What the Model Does Not Guarantee

complete axiomatization universal semantics uniqueness of representation decidability complexity bound empirical validation strict mathematical minimality proof Structure final type universal Stability measure universal tracing algorithm universal Difference codomain

11 Visual Overview of the Minimal Formal Model

11.1 Figure 1: Invariant Core

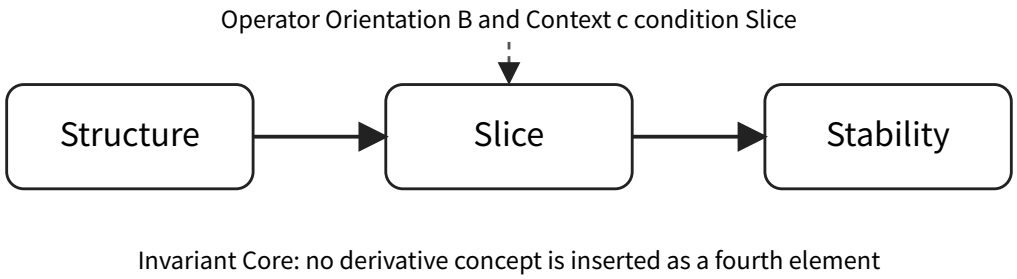


Figure 1: Gyro Logic Core Operator Orientation Context Slice condition additional Core element

Figure 1 theoretical constraint Core Structure → Slice → Stability Operator Orientation Context local articulation Trajectory Difference Boundary Operator Response - Core

11.2 Figure 2: Local Gyro Realization and Context Update

Figure 2

$$g_n = (S_n, B_n, c_n, \Sigma_n, a_n, K_n)$$

Slice process local articulation articulation Stability Scene Incorporated Readability $q_n = \mathrm{Inc}(g_n)$ later readability context Γ_{n+1} update external change e_n

11.3 Figure 3: Contextual Trajectory

$$\mathcal{G}_R = (G, E)$$

$$T_{B,c,\Sigma_T,\Gamma_T} = \mathrm{Trace}_{B,c,\Sigma_T,\Gamma_T}(G, E)$$

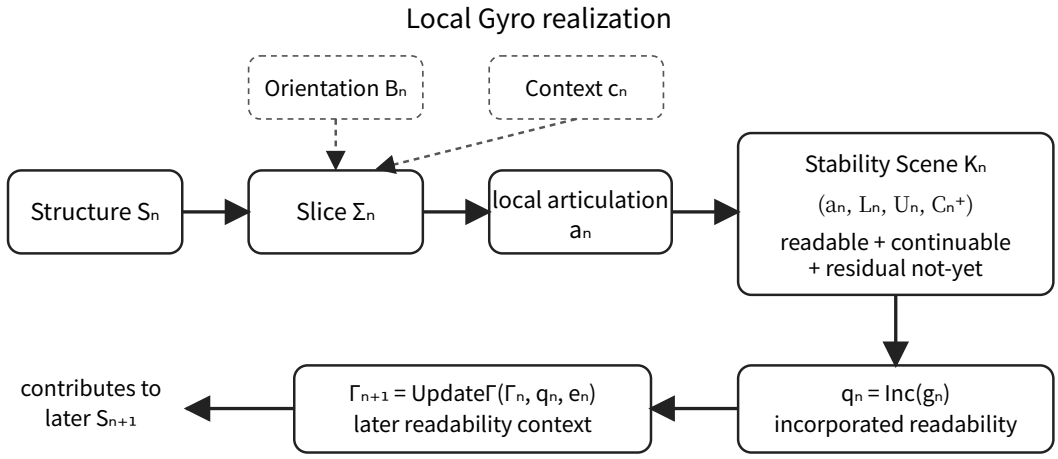


Figure 2: local Gyro realization Stability Scene later readability conditions update

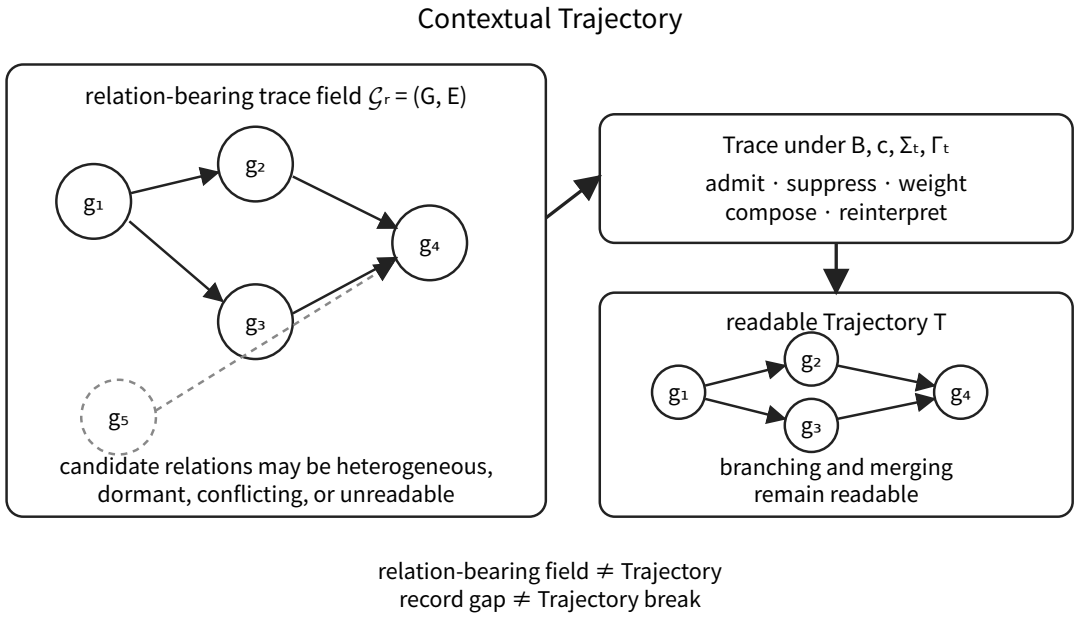


Figure 3: relation-bearing field Trajectory contextual tracing

relation-bearing field heterogeneous / dormant / conflicting / unavailable relations contextual tracing relation admit suppress weight compose interpret resulting Trajectory branch merge gap retrospective revision

11.4 Figure Interpretation Boundary

figures explanatory summary replacement definition Structure box Slice deterministic arrow Stability tuple Trajectory graph

12 Related Work and Formal Positioning

12.1 Relation to the Foundational Gyro Logic Paper

foundational Gyro Logic paper replace formalization companion earlier paper - Core Gyro Logic (Kawakami 2026b) Core Canonical Definition provisional formal schema narrow methodological question

```
foundational paper
=
conceptual introduction and theoretical orientation

present paper
=
minimal formal organization and comparison boundary
```

12.2 Relational and Graph-Based Models

relational structures / graph theory heterogeneous local realizations retained relations graph vertex edge path connectedness branching transformation (Diestel 2017) - graph relevant node / edge already individuated Gyro Logic relation-bearing field Trajectory Trajectory admissibility domain-relative continuity conditions contextual tracing result

12.3 Event Structures and Concurrency

event structures occurrence causal dependency conflict concurrency interleaved sequence (Nielsen, Plotkin, and Winskel 1981; Glabbeek and Plotkin 2009) branching / merging / conflict / partial order / non-linear Trajectory formally represented events enabling / conflict relations local articulation available Slice commitment

12.4 Transition Systems, Model Checking, and Process Algebra

transition systems / model checking states labels transition relations state evolution branching temporal property verification (Baier and Katoen 2008) process algebra interaction concurrency synchronization continuation compositional (Milner 1980, 1982) Gyro Process / Loop / Operator Response / Re-Slice / Defer / Jump implementation Core replacement definition

12.5 Dynamical Systems and Stability

dynamical systems trajectory equilibrium attractor oscillation convergence bifurcation perturbation (Strogatz 2015) Gyro Stability broader ongoing change / residual not-yet Gyro Trajectory time-indexed state solution universally

12.6 Topology, Locality, and Sheaf-Like Structures

topology neighborhood continuity closure separation boundary (Munkres 2000) sheaf theory local information restriction compatibility local-to-global failure (Mac Lane and Moerdijk 1992) Stability Scene locality global non-closure specified underlying space / site / covering universally

12.7 Category Theory and Composition

category theory object morphism composition identity functor heterogeneous transformation (Mac Lane 1998) ordinary morphism domain / codomain presuppose local articulation Slice fully determined codomain

12.8 Belief Revision and Non-Monotonic Context Update

AGM belief revision belief set contraction / revision postulates (Alchourrón, Gärdenfors, and Makinson 1985) Incorporated Readability non-monotonic update Γ deductively closed belief set material / procedural / perceptual / institutional / operational incorporation

12.9 Probabilistic and Statistical Models

probability / statistics uncertainty confidence evidence heterogeneous observation quantify probabilistic graphical model structured dependency / inference framework (Koller and Friedman 2009) graded domain-specific establishment/readability Stability confidence Difference distribution competing Trajectory hypothesis instantiate relevant variable / event / distinction articulable

12.10 Position of the Present Model

reviewed fields formal resources particular objects / relations / spaces / events / operations specified commitment Minimal Formal Model replace distinction coordination role

```
Gyro Logic Minimal Formal Model
≠
a replacement for established mathematics
```

```
Gyro Logic Minimal Formal Model
=
a formal boundary for selecting and coordinating partial models
```

13 Comparison with Existing Mathematical Fields

13.1 Purpose of the Comparison

Gyro Logic field representation assumption Gyro-specific distinction

1. **Representational usefulness** schema effective model
2. **Reduction risk** field universal form

13.2 Relational Structures

heterogeneous objects partial relations admissibility conditions Difference patterns Boundary relations local realization connections numeric / metric assumptions object / relation already available Slice local articulation available

13.3 Graphs and Hypergraphs

$$\mathcal{G}_R = (G, E)$$

trace-bearing relations directed graph multigraph hypergraph branching / merging / competing traces / gaps / retrospective reconnection graph Trajectory

13.4 Order Theory

precedence dependency refinement relevance ordering partial comparability

$$x \preceq_{B,c,\Gamma} y$$

relation domain-relative Difference always orderable

13.5 Topology and Neighborhood Structures

Stability Scene

$$a_n \in N_n$$

neighborhood interpretation Gyro Stability topological stability Boundary topological set boundary not-yet topological openness

13.6 Dynamical Systems

$$x_{t+1} = F(x_t, u_t)$$

model GyroOS / GyroAuth measurable state variable Stability score convergence drift recovery Trajectory state evolution Stability Lyapunov / equilibrium

13.7 Transition Systems and Event Structures

operational succession branching choices enabled actions causality conflict concurrency Gyro Process / Operator Response / Re-Slice / Jump states / events / transitions execution specified pre-individuated Structure model

13.8 Category Theory

heterogeneous object transformation composition identity integration language ordinary morphism domain / codomain fixed Slice articulation universalize

13.9 Logic and Proof Theory

$$\Gamma_n \vdash \varphi$$

proof context Incorporated Readability partial model context extension revision nonmonotonic inference belief revision proposition / predicate / inference rule already articulated

13.10 Constraint Satisfaction and Constraint Propagation

variables domains constraints locally coherent configuration Slice implementation variables / domains / constraints pre-specified Structure universal model

13.11 Probability and Statistics

uncertainty confidence evidence quantify event space / measurable variable already articulated

13.12 Sheaf-Like and Local-to-Global Structures

local data overlap compatibility failure of global gluing local Stability global non-closure base space covering restriction maps universally

13.13 Process Algebra

interaction concurrency communication choice interruption continuation executably / compositionally Gyro Process / Loop action vocabulary / process syntax already articulated

13.14 Comparative Summary

Mathematical field	Strongest Gyro correspondence	Main reduction risk
Relational structures	Difference, Boundary, heterogeneous relations, continuity	Objects and relations appear pre-given
Graphs / hypergraphs	Trace fields, branching, merging, multi-relational connection	Graph mistaken for Trajectory
Order theory	Dependency, relevance, partial precedence	Incomparable Difference forced into order
Topology	Locality, neighborhoods, bounded variation, some Boundary models	Stability reduced to topology; not-yet reduced to openness
Dynamical systems	Evolution, convergence, drift, recovery	Trajectory reduced to state sequence; Stability reduced to equilibrium
Transition / event structures	Branching process, causality, conflict, concurrency	States and events assumed pre-individuated
Category theory	Heterogeneous transformation and composition	Domain and codomain fixed before Slice
Logic / proof theory	Incorporated Readability and context update	Propositions and rules assumed already articulated
Constraint propagation	Emergence of locally coherent articulation	Variables and constraints assumed pre-specified
Probability / statistics	Uncertainty and confidence models	Event space assumed in advance
Sheaf-like structures	Local-to-global compatibility and failure of gluing	Base space and coverings assumed
Process algebra	Operational loops, interaction, response	Action vocabulary assumed articulated

13.15 A Heterogeneous Composite Model

Minimal Formal Model existing disciplines competitor multiple partial models coordinate schema domain-specific implementation relational / hypergraph structures neigh-

borhood / topology logical / non-monotonic context event structure / process algebra probabilistic / dynamical model category-theoretic composition component convenient implementation object Core

13.16 Result of the Comparison

examined field additional assumptions Gyro Logic complete universal model wholly independent mathematics existing fields scope strong partial models

14 Illustrative Examples

example empirical validation uniqueness proof conceptual stress test

14.1 Example 1: Mathematical Problem Solving

proof problem prior assumptions available lemmas notation unresolved obligations Structure S_n Orientation sub-lemma proof invariant isolation goal reformulation Slice

```
Let q_n denote the quantity preserved under the transformation.
```

local articulation available

Stability Scene

$$K_n = (a_n, L_n, U_n, C_n^+)$$

definition intelligible relations remaining proof obligations enabled deductions incorporation

$$q_n = \mathrm{Inc}(g_n), \qquad \Gamma_{n+1} = \mathrm{Update}_\Gamma(\Gamma_n, q_n)$$

log later proof condition

14.2 Example 2: Batter Becoming Cake

batter cooking conditions Structure Slice culinary readiness chemical transformation material continuity product identity Orientation

```
The mixture has set into a cake-like form.
```

articulation pre-existing answer extraction Slice available

$$\mathrm{Id}_q(g_i, g_j) = \text{false}$$

$$\mathrm{CR}(g_i, g_j; B, c, \Sigma, \Gamma) = \text{true}$$

```
identity break
≠
continuity break
```

14.3 Example 3: Authentication Across Changing Conditions

current authentication situation Structure device behavior network time motion observations local articulation

```
The current session is consistent enough
with the previously available user trajectory
to continue provisionally.
```

Stability numerical authentication score score L_n evidential component

Difference

$$\Delta_{B,c,\Sigma}(x) = (\Delta_{\text{device}}, \Delta_{\text{behavior}}, \Delta_{\text{network}}, \Delta_{\text{time}}, \Delta_{\text{motion}})$$

heterogeneous object Boundary Difference tuple ordinary drift suspicious behavior locally usable distinction

14.4 Example 4: Historical Norm Formation

gender equality social recognition society institutions practices conflicts language possible recognition forms Structure legal reform public debate social movement education institutional reinterpretation local articulations

```
Equal treatment is recognized as a legitimate standard in this domain.
```

local Stability Scene enforcement cultural practice exceptions conflicting institutions local not-yet once incorporated equality later law / dispute / interpretation condition

Trajectory chronological event list movements laws decisions institutions practices relation Context admissible / traceable

14.5 Example 5: Missing Data and Trajectory Gaps

sensor system measurement gap Trajectory break before / after realizations admissible relation model constraints material continuity redundant sensors later evidence trace

```
record continuity
≠
Trajectory continuity
```

same event field different contextual tracings support

14.6 Example 6: Search for "All Prefectures Except Kyushu"

query database set difference Gyro Logic negative condition already established classification relative available Slice

```
prefectures that do not satisfy the current Kyushu-membership condition
```

Difference categorical Boundary current Slice usable regional distinction Not Kyushu nothing unknown blank Void collapse

14.7 Cross-Example Observations

examples Structure current observation Slice pre-existing result retrieval local articulation Stability Stability residual not-yet Incorporated Readability stored history Identity / relation existence / traceability / Continuity Readability / Trajectory Difference non-metric Boundary

15 Limitations and Open Problems

15.1 Scope of the Present Model

intentionally limited complete axiomatization universal semantics final mathematical foundation conceptual theory domain-specific implementation

15.2 Provisional Status of Mathematical Types

Structure Slice Stability Context Difference Trajectory universal mathematical type

$$S_n \overset{\Sigma_{B_n, c_n}}{\dashrightarrow} a_n$$

Σ relation / partial map / transition / process object / event / morphism

$$K_n = (a_n, L_n, U_n, C_n^+)$$

every Stability Scene intrinsically four-component tuple claim

15.3 No Proof of Strict Minimality

minimal current theoretical distinctions currently judged necessary formal commitments unique / cardinal / order-theoretic minimality proof

result admissible formal model class canonical concept preservation criterion candidate models ordering / comparison relation component removal distinction loss proof

15.4 Incomplete Semantics of Readability

Readability Stability Definition Incorporated Readability / Continuity Readability important canonical / explanatory language complete universal semantics `Readable(...)` independently validated universal predicate

domain-specific readability binary predicate graded quantity contextual judgment inferential availability relation accessibility structure observer-relative condition heterogeneous family of domain-specific relations

formal condition EstablishedFor_D $\text{CountsAsContinuity}_D$ domain-relative placeholder

15.5 Orientation and Context Are Underspecified

Orientation / Context internal structure fully specified structured state / policy / relation / higher-order constraint Context available conditions / inferential closure / local environment / dynamic structure mutual interaction conflicting Orientations Slice Context change future work

15.6 Boundary Admissibility and Anti-Post-Hoc Limits

`slice-done` local unitization local boundary current Operator judgment Orientation / Context inherited protocol institutional criteria strong event-side transition boundary

flexibility admissibility problem Operator-relativity boundary Orientation / Context boundary inherited boundary after the fact protocol / rule / institution assert claimed provenance independent support

Temporal priority all plausible live alternatives open prior frame constraint present paper methodological anti-post-hoc constraint universal theorem of boundary admissibility universal specificity metric domain-neutral procedure for live candidate boundaries

15.7 Admissibility and Traceability Require Domain Criteria

$$\mathrm{CR}(g_i, g_j; B, c, \Sigma, \Gamma) \iff \exists r(\mathrm{Adm}(r) \wedge \mathrm{Traceable}(r) \wedge \mathrm{CountsAsContinuity}_D(r))$$

admissibility traceability continuity judgment universal criteria domain model permitted relation type evidence conflicting relation handling trace break what counts as continuity uncertainty representation specify

15.8 Retrospective Establishment and Reliability

later Operator available relation trace consequence record later establishment earlier event present establishment

```
past event itself
≠
present establishment about that past event
```

Direct contemporaneous observation every retrospective establishment

```
a remaining trace may support retrospective establishment
≠
a single trace is sufficient to uniquely determine the past event
```

scorch mark lightning arson electrical fault other heat source multiple causes compatible

well-supported retrospective establishment merely plausible story universal reliability criterion Trajectory Re-Slice Incorporated Readability abduction / IBE historical geology forensic reasoning historiographical method open

15.9 Trajectory Reconstruction Is Not Yet Algorithmic

$$\mathcal{G}_R = (G, E)$$

$$T_{B,c,\Sigma_T,\Gamma_T} = \mathrm{Trace}_{B,c,\Sigma_T,\Gamma_T}(G, E)$$

tracing operator search order stopping condition conflict resolution branch selection gap handling uncertainty propagation retrospective revision cost complexity

15.10 Difference Lacks a Universal Codomain

$$\Delta_{B,c,\Sigma} : X \rightharpoonup D$$

heterogeneous codomain different Difference types comparison composition aggregation propagation open

15.11 Stability Has No Universal Evaluation Rule

Stability score local articulation establishment that can continue universal procedure threshold logical satisfaction topological neighborhood invariance robustness confidence interval multi-criteria judgment domain-specific model

15.12 Incorporated Readability Is Not Yet Operationally Identified

$$\Gamma_{n+1} = \mathrm{Update}_{\Gamma}(\Gamma_n, q_n, e_n)$$

q_n realization extract competing incorporated elements reconcile ordinary memory / parameter update empirically

15.13 Empirical Validation Remains Limited

illustrative examples conceptual separability empirical validity explicit dataset / event trace operational definition baseline model measurable success / failure criteria reproducible experiment / simulation GyroOS / GyroAuth PoC success universal theory proof

15.14 Relationship to Existing Mathematics Requires Deeper Study

local articulation partial algebra / event semantics Stability Scene neighborhood / sheaf / domain theory Incorporated Readability non-monotonic logic / belief revision contextual tracing path category / event structure / provenance model heterogeneous Difference enriched relations / order / typed field continuing-event / local-unitization distinction event boundedness / telicity / aspect / process theory rigorous

15.15 Open Problem: Formal Security and Adversarial Conditions

authentication / vulnerability response apply attacker criterion poisoning Context alteration fabricated continuity Difference suppression false Stability security specialization trusted / untrusted evidence adversarial Γ update Boundary manipulation rollback / freeze / defer / review / isolation semantics

15.16 Open Problem: Formal Composition of Local Realizations

$$g_n = (S_n, B_n, c_n, \Sigma_n, a_n, K_n)$$

universal composition operator

$$g_i \circ g_j$$

composition temporal causal logical semantic material contextual

15.17 Open Problem: Criteria for Model Revision

formal model provisional canonical definition conflict distinction collapse unnecessary ontological assumptions important domains fail theoretical benefit implementation observable / inferential evidence revision

15.18 Summary of Limitations

present model

- final ontology of Structure
- universal mathematical type for Slice
- complete universal semantics of readability
- universal Stability metric
- universal boundary-admissibility rule
- universal Difference codomain
- executable tracing algorithm

- general reliability criterion for retrospective establishment
- proof of strict minimality
- complete security model
- empirical validation across domains

distinction reduction currently unjustified component future development disciplined formal boundary

16 Conclusion

Core

```
Structure
↓
Slice
↓
Stability
```

Gyro Logic exploratory Minimal Formal Model Canonical Definition equation Gyro Logic theoretical distinctions theory commitment compact / internally consistent formal schema *minimal* exploratory operational sense strict mathematical minimality

Structure one universal mathematical object type Slice Slice process local articulation `slice-done` underlying event objective end local unitization Stability unitization articulation establishment structured local scene residual local not-yet

Canonical term `readable` universal independent `Readable(...)` predicate domain-specific formal model operational precision establishment continuity admissibility tracing justified conditions

Incorporated Readability stored history later conditions potentially non-monotonic update Continuity Readability Identity Trajectory state sequence log event accumulation local Gyro realizations admissible relations contextual tracing Difference distance numerical error Boundary Boundary derivative locally usable distinction

integrated local realization

$$g_n = (S_n, B_n, c_n, \Sigma_n, a_n, K_n)$$

Core-relative relation

$$S_n \xRightarrow{\Sigma_{B_n, c_n}} a_n \xRightarrow{\text{Stab}} K_n$$

incorporated contribution

$$q_n = \text{Inc}(g_n)$$

$$\Gamma_{n+1} = \text{Update}_\Gamma(\Gamma_n, q_n, e_n)$$

later Structure conditions

$$(S_n, \Gamma_{n+1}, e_n) \nrightarrow S_{n+1}$$

Continuity Readability admissible / traceable relation domain-relative continuity condition characterize Trajectory relation-bearing field contextual tracing operation Difference weakly typed partial heterogeneous mapping

```
Slice process
≠
local articulation / local unitization
≠
Stability
```

```
stored history
≠
Incorporated Readability
```

```
Identity
≠
Continuity Readability
```

```
relation field
≠
Trajectory
```

```
Difference
≠
Distance
≠
Error
≠
Boundary
```

existing mathematical fields relational structures graphs topology dynamical systems transition systems category theory proof theory constraint propagation process algebra useful partial model Gyro Logic additional assumptions capture position mathematical isolation premature reduction domain-specific mathematical models explicit assumptions heterogeneous formal organization

introductory Gyro Logic paper formalization companion introductory work Gyro Logic current conceptual distinctions mathematical comparison validation later implementation minimally organize

future research domain-specific semantics of establishment/readability admissibility traceability boundary provenance / admissibility local-unitization event/process theories retrospective establishment evidentiary / abductive methodologies local realization composition executable / simulation-based instantiation non-monotonic Incorporated Readability adversarial update / criterion poisoning

present result deliberately limited proposed schema uniquely minimal empirically valid across domains computationally decidable complete universal boundary / retrospective-reliability criteria modest foundation Gyro Logic Core central distinctions narrow pre-existing mathematical forms collapse disciplined formal organization

Declarations

Conflict of Interest

Funding

Data Availability

empirical dataset

Use of Generative AI and AI-Assisted Tools

OpenAI ChatGPT Anthropic Claude Claude Code Google Gemini AI manuscript content theoretical claims citations references final text review verify edit

Code and Materials Availability

manuscript sources figures assembly scripts PDF-generation workflow validation scripts Gyro Logic repository : https://github.com/gitGyro-Dev/gyrologic

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